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Angular Asymmetry in Dimuon Production

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ABSTRACT

The Large Electron-Positron Collider (LEP) at CERN tested the electroweak theory of the Standard Model through its estimation of parameters according to the differential cross-section of dimuon production. By using Monte Carlo rejection sampling, this project simulates dimuon event generation under realistic conditions at the LEP, including finite detector acceptance and angular resolution constraints. The parameters of the cross-section equation are estimated by considering the forward-backward asymmetry, a simple quadratic regression model coded from scratch, a curve-fitting module from the Python package SciPy, and by minimising the χ^2-statistic across various configurations. Results show that parameter uncertainty scales according to a power law in the number of events generated, with angular smearing significantly influencing estimated parameter values, and the χ^2 method proving most robust for accurate estimation. These findings illustrate the trade-offs between statistical precision, detector performance and methods of analysis when determining parameters from real-world apparatus.

1 INTRODUCTION

The LEP remains the most powerful accelerator of leptons ever built, containing four large detectors for experiments on lepton collisions [1]. However, like many such accelerators before and since, it was still subject to physical constraints as to the accuracy of its measurements. In an ideal collider, a collision between particle beams would be completely surrounded by a detector, covering the full solid angle, and with perfect resolution of trajectory angles. In reality, the beam pipe, power and cooling cables, and other equipment, such as for shielding, will obstruct the detectors ability to record events across the full solid angle. Additionally, electronic noise, pixel size or particle wavelengths can reduce the resolution of the detector, smearing the recorded deflection angles.

The consequence of this is that measurements made by these colliders come with a degree of uncertainty attached. These uncertainties can stack up if not properly accounted for and yield unsuitable results, given that the essence of precision is a key part of particle physics. However, there are methods of increasing the reliability of collider results which goes beyond fine-tuning highly precise equipment, such as increasing the number of production events within the detector. This reliability is crucial for measuring precise angular distributions, which describe fundamental parameters of particle interactions. It's on this basis that this project is based by simulating the production of muon pairs at colliders like the LEP and testing how the number of events, and practical detector conditions, influence the precision of parameters.

2 THEORY

The LEP initially started with an energy of ∼91 GeV, peaking at 209 GeV [2], for the purpose of producing Z-Bosons. The Z-and-W-Bosons are responsible for transmitting the Weak force in particle physics and from the creation, and subsequent decay, can be used to test the Standard Model and Electroweak theory. To do this, the LEP would collide one beam of electrons with one beam of positrons, each beam with four 'bunches' of the particle, collided precisely at the locations of the four detectors. Typically, bunches were a centimetre long and had a forward-facing area of one-square millimetre [3]. Specifically for this project, we are concerned with

$$e^- e^+ \rightarrow \mu^- \mu^+ \tag{1}$$

as the reaction producing Z-bosons.

One of the key applications of the collisions is measuring the differential cross-section for dimuon production. The differential cross section quantifies the number of particles emerging from a solid angle, denoted as $d\Omega$. This allows us to describe the probability of particles scattering into a specific direction [4 5]. The differential cross-section considered in this project takes the form:

$$\frac{d\sigma}{d\cos(\theta)} \propto 1 \; + \; a_1 \cos(\theta) \; + \; a_2 \cos^2(\theta) \tag{2}$$

where θ is the angle between the incoming electron and the outgoing μ^+, σ is the cross-section and a_1, a_2 are parameters to be estimated.

There are several ways the parameters can be estimated, four of which will be covered in this project. But first, we discuss how the events are to be treated.

2.1 Monte Carlo Sampling

One should recognise that the angle of scattering for μ^+ is, for all intents and purposes, totally random, and we'll assume it

does not favour one scattering angle over another. To mimic this randomness, a Monte Carlo rejection sampling process can be used to generate random values of $\cos(\theta)$ across the defined range. By then applying a binned fit, where bins have some defined width which covers a small range of cosine values, a histogram can be plotted which mimics the differential cross-section function. The area of this binned fit corresponds to the number of events generated, and from it N_- and N_+ can be inferred.

2.2 Parameter Estimation Techniques

The first is by considering the forward-backward asymmetry of detected events. Since θ denotes the scattering angle, it can range from [0, π] radians (π being the maximum since this is perfect back-scattering), granted the detector has perfect acceptance with no impediments on the available solid angle. This means that $\cos(\theta)$ in an ideal detector has a range between $[-1.0,\ 1.0]$. By recording the events detected and their deflection angle, the forward-backward asymmetry can be described as the fractional difference between those events with $\cos(\theta) > 0$ and $\cos(\theta) < 0$:

$$A_{FB} = \frac{N_+ - N_-}{N_+ + N_-} \tag{3}$$

where N denotes the number of events with positive/negative $\cos(\theta)$. Then, 2 can be integrated between the minimum and maximum values that $\cos(\theta)$ can take in the system we're modelling. For generalisation purposes, we assume $\cos(\theta) \in [-c\ , c]$, and by integrating between $[-c, 0]$ for N_- and between $[0, c]$ for N_+, we arrive at the following:

$$a_1 = 2\left(\frac{1}{c} + \frac{c}{3}a_2\right)A_{FB} \tag{4}$$

where for full acceptance we have $c = 1.0$. The significance of the $a_1 \cos(\theta)$ term originates from the quantum interference between the electromagnetic γ and weak Z^0 interaction channels, breaking the forward–backward symmetry. The term a_2 can be estimated from the total cross section, meaning a_1 can be inferred from the asymmetry.

The second is via a simple quadratic-regression machine learning model. The premise is that we start with a function mimicking the target distribution we're looking for, with randomly initialised coefficients. Then, evaluate this function at chosen $\cos(\theta)$ values, comparing the result to what those values should correspond to on the true target distribution; depending on how different the results are, change those coefficients accordingly. This method requires choosing an appropriate loss function as well as other hyper-parameters, such as the learning rate. Letting $L(\hat{y}, y)$ be an arbitrary loss function for the predicted and actual values, each parameter, α, is updated across n_{iter}-iterations according to the following gradient descent function:

$$\alpha_t = \alpha_{t-1} - \eta \frac{\partial L}{\partial \alpha}\Big|_{\alpha=\alpha_{t-1}} \tag{5}$$

where t is the current iteration and η is the learning rate. The advantage of such a method is, despite its conceptual simplicity, it often yields surprising accuracy, however, it remains a tool more for preliminary fitting due to no uncertainty definition and a heavy dependency on hyper-parameter fine-tuning which is often the result of trial and error[6].

The third method is using a curve-fitting module from the SciPy library in Python[7]. The 'curve_fit' function employs a non-linear least squares approach to optimise the parameters by minimising the sum of squared residuals between the model and the data. Internally, it interpolates between gradient descent and other methods for stability. The module returns not only the optimised coefficients but also an estimate of their covariance matrix, from which parameter uncertainties can be derived as the square-rooted diagonal. This built-in uncertainty makes it a practical choice for fitting, though it still may struggle with poor initial guesses or highly correlated parameters.

The fourth method chosen was a χ^2 minimisation approach. It is similar in nature to the curve-fitting function, in that it compares the squared-residuals between expectation and what's observed/predicted in the context of a binned fit of the data. The method is based on finding the conditions to minimise this quantity, meaning we look for the parameters of the differential cross-section such that it most closely mimics the true data. χ^2 is a decimal value for each configuration of a_1, a_2 parameters and takes the form:

$$\chi^2 = \sum_i^{\text{bins}} \frac{(O_i - E_i)^2}{E_i} \tag{6}$$

where O_i is the observed bin frequency for bin 'i' and E_i is the expected bin frequency.

Some of the fundamental statistical assumptions made throughout are: (1) event generation is totally random, (2) the angular resolution smearing can be modelled as Gaussian noise, (3) the cross-section functional form is known exactly, and (4) Monte Carlo sampling provides an unbiased estimator of the true distribution.

3 METHODS

Vectorised Sampling For Monte Carlo sampling an initial sequential approach was implemented, performing the rejection/acceptance one value at a time, but this was later changed for a vectorised approach, processing $\cos(\theta)$ values in batches drawn from a uniform distribution. Whilst this approach improved computational time it did mean that values shared a generating seed, reducing the degree of randomness in generating the events. However, given the large sample sizes processed in batches (typically $N \geq 10^4$), this reduction in seed independence was assumed to have negligible effect on statistical interpretation under the assumption of pseudo-randomness inherent to Monte Carlo simulations. Batch size was set to $2N$ to ensure sufficient candidates per iteration, with final accepted values cropped to the exact N required to mimic independent event generation.

Ideal and Realistic Detector The simulated detector was considered in two contexts: ideal, with perfect acceptance

and angular resolution, and one with restricted acceptance and 'smearing' effects on angular measurement. In the former case, this meant using $|\cos(\theta)| \leq 1$. In the latter, detector acceptance meant $|\cos(\theta)| \leq 0.9$. An angular resolution of 3 mrad was assumed and accounted for by drawing events within the restricted acceptance then evaluating the theoretical angle resulting in the scattering by adding Gaussian noise $\theta_{\text{noise}} \sim \mathcal{N}(0,\ 3 \times 10^{-3})$ according to the resolution. Smeared angles were clipped to $[0, \pi]$, then transformed back to $\cos(\theta)$ before re-applying the acceptance cut.

Forward-Backward Asymmetry The ground-truth values for a_1, a_2 were set to 0.04 and 1.0 respectively, mimicking realistic values as measured by LEP. The ground-truth for a_1 was hidden when using the asymmetry to infer its estimate and uncertainty according to 4. In practice, the observed N_+ and N_- were counted from the Monte Carlo sample, and A_{FB} was computed via 3. The parameter a_2 was treated as known—estimated from total cross-section—allowing a_1 to be inferred directly. To quantify uncertainty, the process was repeated over $M = 500$ independent samples, with the standard deviation of the resulting a_1 estimates taken as the statistical error.

However, in most practical cases considered in the project a random initialisation of the parameters was chosen. This was done to ensure results and interpretations were independent of specific configurations, as all processing techniques should be stand-alone.

Gradient Descent For the quadratic-regression model, a learning rate of $\eta = 0.01$ was chosen for $n_{\text{iter}} = 1 \times 10^4$ iterations. The loss function used was a mean-squared error (MSE). Parameter updates followed the gradient descent rule given in 5, with initial coefficients drawn uniformly from $[-1, 1]$. Although this approach converges reliably for the simple quadratic form, it offers no inherent uncertainty estimates—a limitation noted in the discussion.

SciPy Curve Fitting The curve_fit function from SciPy was applied to binned histogram data, with bin centres x_i and heights y_i scaled by the estimated integral area. Minimisation of the sum of squared residuals returned the optimal parameters a_1, a_2 and a covariance matrix. Uncertainties were taken as the square root of the diagonal entries. Parameter bounds were set to $[-1, 1]$ for both a_1 and a_2 to reflect physical limits.

χ^2 Method A binned χ^2 approach was implemented by comparing observed bin counts O_i to expected counts E_i computed from the theoretical cross-section for trial (a_1, a_2) from the ideal detector. The expected count for bin i was:

$$E_i = \frac{f(x_i; a_1, a_2) \times \text{Bin Width} \times \text{N-events}}{\text{Integral Area}} \tag{7}$$

where f is the target function from 2 and the integral normalises the distribution over the acceptance region. A 2D grid search over $a_1, a_2 \in [-1, 1]$ was performed to locate the χ^2 minimum. Parameter uncertainties were extracted from the contour where $\chi^2 = \chi^2_{\min} + 2$, corresponding approximately to the 68% confidence region for two parameters. Coverage was tested over 100 independent trials to assess statistical reliability.

Uncertainty and Validation Statistical uncertainties were propagated analytically for the forward-backward method and derived from covariance or contour analysis for curve-fitting and χ^2 methods. Validation was performed by comparing Monte Carlo integral estimates with analytical results, and by checking recovered A_{FB} values against the published LEP measurement 0.0167 ± 0.0013 for the $(a_1, a_2) = (0.04, 1.0)$ benchmark[8].

4 RESULTS

The Monte Carlo sampling was first tested with trial values of $a_1 = a_2 = 1.0$ to evaluate its ability to reconstruct 2. The estimated area was found to be 2.296 and deemed acceptable to the true integral of 2.286. This also revealed reliability in the method for inferring the number of events contained either side of $\cos(\theta) = 0$.

Random values of the parameters were produced which remained constant throughout (apart from LEP comparison) at $a_1 \approx 0.526$ and $a_2 \approx 0.640$. The first forward-backward asymmetry method used the ideal case, which resulted in an estimate of $a_{1,\text{ideal}} = 0.528 \pm 0.024$. By repeating these measurements across different a_1 values, keeping a_2 fixed, a plot was produced similar to 1 as shown for the smeared scenario. The smeared forward-backward test also produced a precise value of $a_{1,\text{smeared}} = 0.527 \pm 0.025$, of which the a_1 Vs A_{FB} trend is shown in 1.

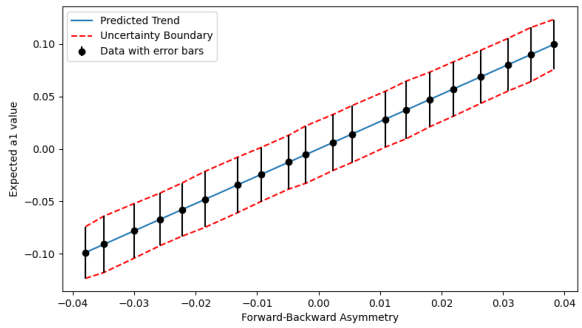

Fig. 1: Expected a1 value as a function of forward-backward asymmetry for the smeared detector system.

The absolute and relative uncertainties on a_1 estimates for both the ideal and smeared case were measured with the number of events increasing to 10000. The results are shown in 2, which demonstrates that the absolute error scales as a power law in number of events.

The validity of the model was then tested against the LEP data for A_{FB} for hidden $a_1 = 0.04$ and $a_2 = 1.0$ using

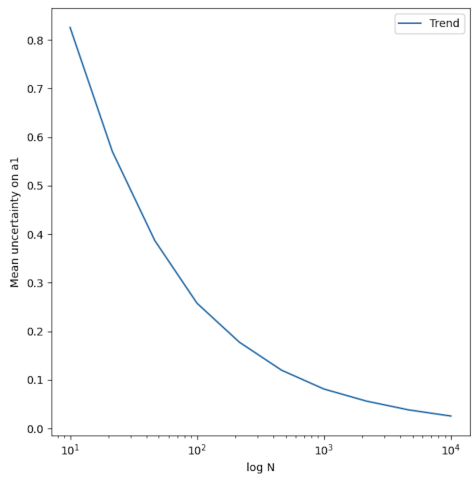

Fig. 2: Absolute error of the smeared detector as a function of log(N-events). Demonstrates an inverse power law.

both the ideal case and the smeared version of 4 with $c = 0.9$. An ideal LEP detector was theorised to have $a_1^{\text{ideal LEP}} = 0.0445 \pm 0.0035$ whereas a smeared LEP had $a_1^{\text{smeared LEP}} = 0.0471 \pm 0.0037$. In both cases the ground-truth a_1 was not contained within the uncertainty.

Returning to the randomly initialised parameters, the quadratic-regression model, after 10000 iterations the model recorded a loss of $L = 0.0038$. SciPy's curve-fitting package also yielded accurate predicted values, with the true parameters lying within its uncertainties. Both methods have their results detailed in 1.

The χ^2 test, after cycling through a (200×200) grid of (a_1, a_2) trial values managed to successfully contain the true parameters within its 1σ uncertainty 67% of the time, compared to the 68% theoretical coverage. A sample estimate for the parameters can be found in 1 and below is a figure of the landscape searched.

Then the mean absolute error was found for a_1 to relate the trend back to earlier results. This time one parameter was fixed and the other allowed to vary from $[-1, 1]$. The result was that event counts less than $N = 1000$ the error is almost scochastic in nature and unpredictable, however beyond this limit all trends (meaning all separate cases where one parameter has been varied) begin to decrease in error simultaneously. The result is shown in 4.

The final result was collecting the four methods' parameter estimations and uncertainties into a table, shown in 1

5 DISCUSSION

All four methods successfully recovered the underlying parameters to within a reasonable statistical margin. The forward–backward asymmetry, while simple and intuitive, proved robust even when detector smearing was introduced; its uncertainty increased only marginally, confirming that the

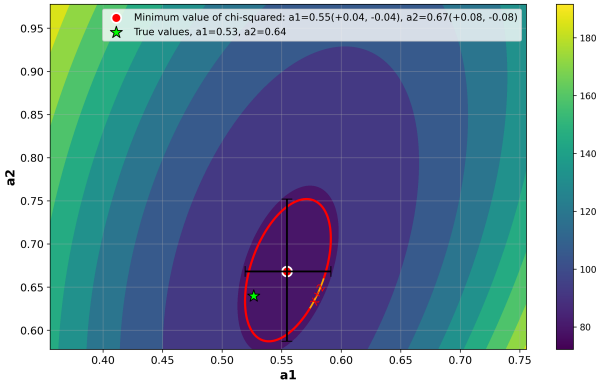

Fig. 3: Contour plot of χ^2 values with the minimum value being the estimated configuration for the true parameters (a_1, a_2)

method is relatively insensitive to angular resolution effects at the 3 mrad level considered.

The quadratic-regression model converged reliably but lacks an intrinsic uncertainty estimate, limiting its use for parameter extraction. Its value lies more in providing a quick initial fit than in delivering results.

SciPy's curve-fit delivered accurate estimates with reasonable uncertainties, and its results aligned closely with the quadratic-regression output, validating both approaches. The fact that its uncertainties contained the true parameters in most trials demonstrates its reliability for this type of quadratic fitting problem, whereas if the fit had been less quadratic in nature it may have struggled.

The χ^2 minimisation approach provided the most complete statistical picture, yielding a two-dimensional confidence region and achieving near-perfect coverage (67 $\%$ against a theoretical 68 $\%$) once the contour level was set to $\chi^2_{\min} + 2$. This adjustment was necessary because, in two dimensions, the 1σ region corresponds to $\Delta\chi^2 = 2.3$ rather than 1[9 10].

The scaling of parameter uncertainties with event count followed a clear inverse-power-law behaviour in both ideal and smeared scenarios. This demonstrates that, even with realistic detector limitations, increasing the statistics remains the most straightforward way to improve precision. However, the χ^2 analysis revealed that for small samples ($N < 10^3$) the error estimates become erratic, revealing the need for a sufficient Monte-Carlo procedure before reliable uncertainties can be taken.

When tested against the LEP benchmark ($a_1 = 0.04$, $a_2 = 1.0$), the inferred a_1 values from the forward–backward method lay just outside the quoted experimental uncertainty. This small difference may reflect several reasons: the Gaussian smearing model or assuming perfect a_2 knowledge for instance. In a reality, a_2 would also carry an uncertainty that would influence a_1.

There remains two concerns: one around the reproducibility of results and another around the vectorised sampling approach. For the former, it appears that the randomness can have more

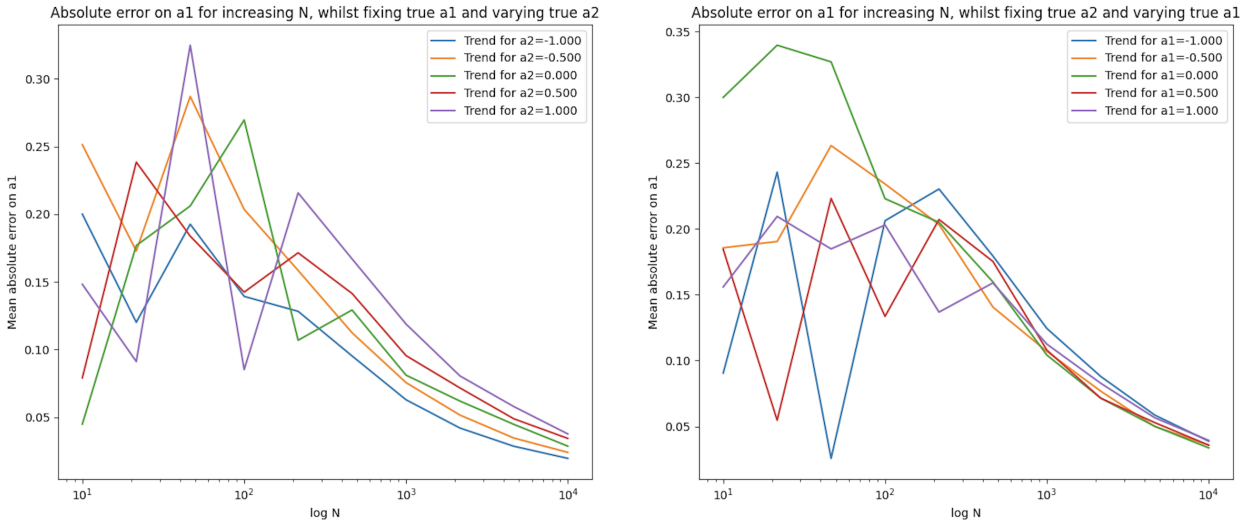

Fig. 4: Left: Absolute error on a_1 for varying a_2. Right: Absolute error on a_1 for varying a_1.

Table 1. Comparison of parameter estimation methods for a_1 and a_2 with $N = 10^4$ events. The true parameters are $a_1 \approx 0.526$ and $a_2 \approx 0.640$.

Method	Estimated a_1 ($N = 10^4$)	Absolute a_1 Uncertainty	$(a_1^{\text{true}} - a_1^{\text{est}})$	Estimated a_2 ($N = 10^4$)	Absolute a_2 Uncertainty	$(a_2^{\text{true}} - a_2^{\text{est}})$
Ideal F-B	0.5276	0.0244	-0.0011	—	—	—
Smeared F-B	0.5267	0.0251	-0.0002	—	—	—
Quad-Reg	0.5372	—	-0.0107	0.6234	—	0.0164
CurveFit	0.5372	0.0228	-0.0107	0.6234	0.0294	0.0164
Chi-Squared	0.5546	$[-0.0353, 0.0368]$	-0.0281	0.6682	$[-0.0811, 0.0838]$	-0.0284

dramatic effects for certain configurations of the randomly initialised parameters, such as changing the χ^2 coverage each time its initiated. For the latter, whilst it offered significant savings for computation, the sharing of random seeds between batches could, in principle, introduce correlations. For the sample sizes used here ($N \geq 10^4$), this effect is assumed negligible, but for extremely precision-critical applications, it might be best to revert to sequential sampling and look to improve efficiency in other areas, perhaps hardware or programming language.

6 CONCLUSION

The project untaken has demonstrated how Monte-Carlo rejection sampling can simulate dimuon production at a collider such as LEP, and how several common parameter-estimation techniques perform under both ideal and realistic detector conditions. Most methods successfully recovered the input parameters within expected statistical fluctuations, with the χ^2 approach providing the uncertainty measurement most reliable through its 2-D confidence regions.

The results confirm that detector effects, such as finite acceptance and angular resolution, do affect precision, but that it is moderate and can be largely countered by collecting larger event samples. The forward-backward asymmetry method proved robust to smearing, making it a practical choice when a_2 is well-known from independent measurements. Meanwhile, the curve-fitting and χ^2 methods offer more complete parameter estimation when both coefficients must be determined.

The work illustrates the trade-offs between computing efficiency, detector performance and methods of analysis that are common in experimental particle physics. For future work, the simulation could be extended to include other uncertainties such as efficiency variations, or use a more complex models instead of simple Gaussian smearing.

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